Preconception α-cypermethrin exposure alters gene expression in the embryonic mouse brain

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Background

Pyrethroid insecticides such as α-cypermethrin are widely used, leading to continuous human exposure through food and the environment. Increasing evidence suggests that insecticide exposure during sensitive developmental periods may affect brain development and increase the risk of neurodevelopmental disorders; however, the underlying molecular mechanisms, particularly following preconception exposure, remain unclear. This study aimed to investigate the effects of preconception α-cypermethrin exposure at 0.3 mg/kg on gene expression and biological pathways in the embryonic brain across maternal, paternal, and combined parental groups using RNA-sequencing.

Methods

RNA-seq data from 24 C57BL/6J mouse embryonic forebrain samples were analysed. Samples were grouped into control, maternal, paternal, and combined parental exposure groups, and differential expression was evaluated by comparing each exposure group to the control. The data were processed using a Snakemake workflow. Raw sequencing reads were quality-checked using FastQC and trimmed with fastp to remove adapters and low-quality bases. Filtered reads were aligned to the *Mus musculus* reference genome (GRCm39) using HISAT2, and gene expression levels were quantified with featureCounts using mouse gene annotation (GTF). Quality metrics from all steps were summarised using MultiQC. Differential gene expression analysis was performed in R using DESeq2, applying normalisation, the Wald statistical test, and false discovery rate (FDR) correction to identify significantly regulated genes between exposed and control groups.

Conclusions

Preconception low-dose α-cypermethrin exposure altered embryonic forebrain transcriptomics in a parent-specific manner. Maternal exposure was primarily associated with cell cycle and chromosome segregation processes, paternal exposure with ribosomal and mitochondrial activity, and combined parental exposure with broader mitochondrial and oxidative phosphorylation signatures. Across all exposure groups, neuronal and synaptic pathways were consistently downregulated, indicating a shared neurodevelopmental disruption pattern.

Contribution

- Bikulciene I. - conducted RNA-seq data processing, differential gene expression analysis, and poster presentation.
- Malik A. - performed RNA-seq data processing and differential gene expression analysis for independent verification.
- Rimas V. - compiled results and conclusions to poster presentation, worked on poster design.

Results

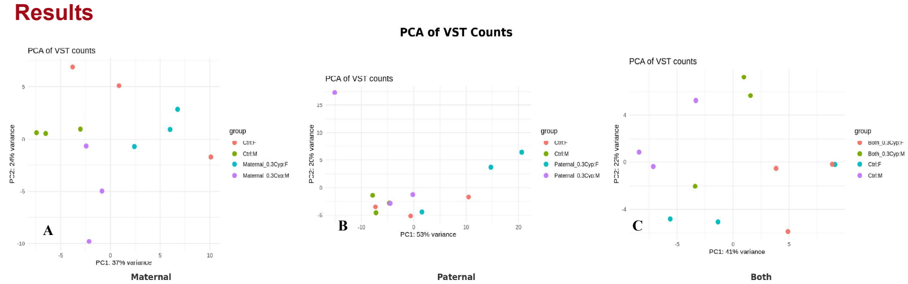

Figure 1. PCA of VST gene expression data. Sample clustering for (A) maternal, (B) paternal, and (C) combined parental exposure vs control groups indicates global transcriptomic differences.

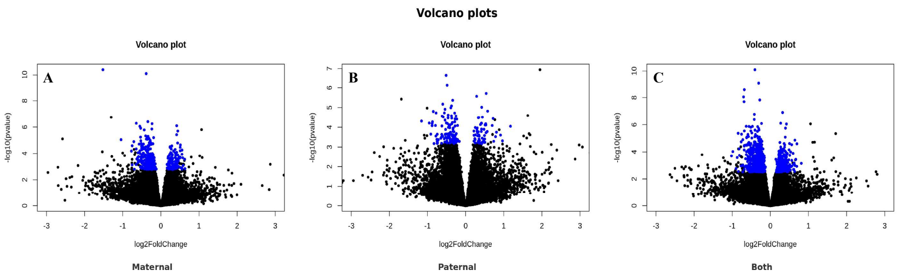

Figure 2. Volcano plots of differential gene expression. Differential expression between control and (A) maternal, (B) paternal, and (C) combined parental exposure groups is shown. Significantly differentially expressed genes (FDR-adjusted) are highlighted.

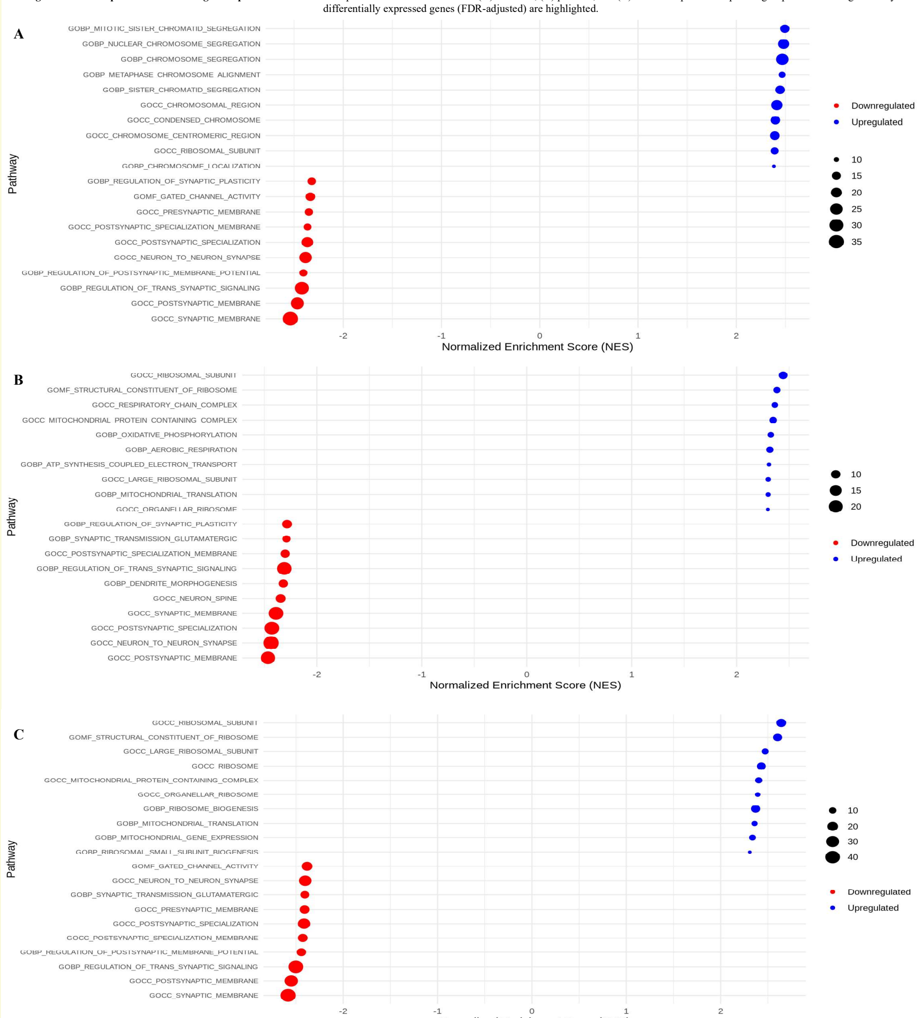

Figure 3. Pathway enrichment analysis. (A) Maternal: translation-related pathways; (B) paternal: cell cycle/DNA damage; (C) combined: mitochondrial pathways; synaptic processes are downregulated across groups.